Arjun Pavanje

ee24btech11005@iith.ac.in | arjupavanje@gmail.com | +91 6362258560 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.76
Narayana Olympiad SCH Sahakarnagar Bangalore	Class XII (CBSE)	2024	95.60%
CMR National Public School Bangalore	Class X (CBSE)	2022	96.60%

SCHOLASTIC ACHIEVEMENTS

Received **A+** (outstanding performance) in the course Electronic Devices and Circuits Lab where I built a short range working FM transmitter and receiver.

PROJECTS

5-Stage Pipelined RISC-V Processor (Github):

- Designed and implemented a **5-stage pipelined RV64IM** RISC-V core from scratch in Verilog, including the datapath, control unit, and hazard/forwarding logic
- Verified full-core functionality through **Vivado** simulation across the RV64IM instruction set; FPGA deployment in progress
- **Parameterized core** architecture that is configurable between 32-bit and 64-bit (RV32IM / RV64IM) mode

Floating Point Unit (FPU), Hazard Detection Unit (Github):

- Designed and Implemented Floating Point and Hazard Detection Unit (Data Hazards primarily) for a 5-stage pipelined RV64IMFD processor
- Designed units ended up being implemented on PYNQ Z2 FPGA
- Developed a custom approximate division algorithm using fast inverse square root to conserve FPGA LUTs

ESP32 based WiFi attacks (Github) (Demo):

- Developed an ESP32-based Wi-Fi security research tool implementing IEEE 802.11 management frame injection for 2.4 GHz networks using the ESP32's raw packet transmission capabilities
- Implemented **deauthentication** and **beacon spam** attack modules to demonstrate wireless security vulnerabilities by disconnecting clients and broadcasting multiple spoofed SSIDs in controlled testing environments
- Designed a **web-based control interface** hosted on the ESP32, enabling users to configure targets, launch attacks, and monitor the tool remotely through a browser

RF Transceiver PCB design:

- Designed the digital section of a mixed-signal RF transceiver PCB in Altium Designer, incorporating an STM32 microcontroller and SD card interface
- Created the schematic and PCB layout following datasheet-recommended routing, grounding, and decoupling guidelines to maintain signal integrity and reduce coupling into sensitive RF circuits

MINI PROJECTS

Eigenvalue calculator (Github):

- Designed and implemented an algorithm to calculate eigenvalues of a matrix (complex entries supported) in C
- Algorithm involves **Householder Reflection** to bring the matrix to Upper-Hessenberg form, and then **Givens Rotation** to perform QR Decomposition
- Improved convergence with **Rayleigh Shifts**

Pipelined Radix-2 8 point FFT accelerator (Github):

- Designed and implemented 3-stage pipelined 8-point Radix-2 Decimation-in-Frequency (DIF) Fast Fourier Transform (FFT) accelerator in Verilog
- Design implements a streaming Single-Path Delay Feedback (SDF) architecture, capable of accepting one complex input sample every clock cycle while sustaining a throughput of one complex output sample per clock after the pipeline is filled
- Implemented fixed point system for computations, reorder buffer, butterfly and twiddle factor units

Digital Clock using Assembly (Github):

- A fully functional HH:MM:SS digital clock implemented on the ATmega328P (Arduino Uno) using AVR Assembly language, six 7-segment displays, and only 1 7447 BCD-to-7-segment decoder
- Implements time incrementation using Karnaugh map (K-map) derived logic

Scientific Calculator using AVR-GCC (Github):

- Designed and implemented a fully functional scientific calculator built on the ATmega328P (Arduino Uno) using Embedded C (avr-gcc)
- Mathematical functions (like trigonometric, logarithm, exponent) were implemented using numerical methods (RK-4) rather than using standard libraries
- The calculator features a custom-built button matrix, an LCD-based user interface, and support for advanced scientific operations

Pipelined FIR Filter (Github):

- Designed and implemented 3-architectures of parameterized, pipelined FIR filter in verilog - direct, optimized (symmetric), genvar-based

SKILLS

Languages: C, Python, Verilog, Assembly

Tools: Git, Vim, Linux, Latex, LTSpice, Altium

POSITIONS OF RESPONSIBILITY

Core at Kludge, the Information Security and Networking Club of IITH

RELEVANT COURSES

Computer Architecture (CS2323), Signal Processing (EE2000), Machine Learning (EE2802), Digital Systems (Theory - EE1201, Lab - EE1501), Data Structures and Applications (ID2230), Semiconductor Device Fundamentals (EE2104)